

Parallel Protein Sequence - Structure Feature Extraction and CATH Classification

Project Overview

This project classifies proteins using both their amino-acid sequence and 3D structural information.

The project combines:

- Protein sequence information using ESM-2
- 3D structural information using 28 structural features
- Machine learning for CATH Class classification
- HPC techniques using Sequential CPU, OpenMP, and CUDA

Overall workflow:

TEDBench → Sequence + 3D Structure → Feature Extraction → Feature Fusion → ML Classifier → CATH Class

Main Research Question

Does combining protein sequence information with 3D structural information improve protein classification, and how can HPC techniques accelerate the computation?

Dataset

We use the TEDBench/ted dataset.

The dataset provides:

- Protein sequence
- 3D backbone coordinates
- pLDDT values
- Residue information

- CATH-related labels

The current classification task is CATH Class classification with four classes:

1, 2, 3, 4

Sequence Feature Extraction Using ESM-2

We use the pretrained protein language model:

ESM-2: [facebook/esm2_t6_8M_UR50D](#)

The amino-acid sequence of each protein is given to ESM-2.

ESM-2 converts the sequence into a numerical embedding containing:

320 sequence features

These embeddings represent information learned from the protein sequence and are used as input to the classifier.

Structural Feature Extraction

The TEDBench dataset provides 3D backbone coordinates for each protein.

We process these coordinates to calculate 28 numerical structural features.

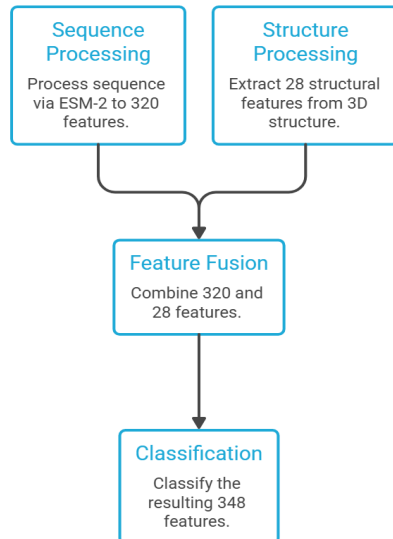
These features describe properties such as:

- Protein compactness
- Radius of gyration
- End-to-end distance
- C α distance statistics
- Pairwise distance statistics
- Contact-related properties
- Spatial distribution of the protein structure

Thus:

3D coordinates → Structural processing → 28 structural features

Feature Fusion



The sequence and structural features are combined into a single feature vector.

Machine Learning Classifier

The combined 348-dimensional feature vector is given to a machine-learning classifier.

The classifier predicts one of the four CATH Classes:

Class 1, Class 2, Class 3, Class 4

The model is trained using the training set, monitored using the validation set, and evaluated using the test set.

HPC Part

We compare three computational approaches:

Sequential CPU

- 1 CPU thread
- Used as the baseline

OpenMP CPU

- 2 threads
- 4 threads
- 8 threads
- 12 threads

CUDA

- NVIDIA Tesla T4 GPU

The comparison focuses on:

- Execution time
- Speedup
- Efficiency
- Classification accuracy
- Feature extraction performance

Consolidated HPC Classification Results

Method	Thread s/GPU	Classifier Training (s)	Classificat ion Time (s)	Speed up	Efficiency	Test Accur acy	Balanced Accuracy	Macro F1	Weighted F1
Sequential CPU	1	8.313	8.388	1.00×	100.0%	0.8053	0.6773	0.6235	0.8301
OpenMP CPU	2	5.710	5.733	1.46×	73.15%	0.8053	0.6774	0.6233	0.8235
OpenMP CPU	4	4.101	4.119	2.04×	50.91%	0.8047	0.6769	0.6229	0.8227
OpenMP CPU	8	3.007	3.026	2.77×	34.65%	0.8053	0.6774	0.6233	0.8234
OpenMP CPU	12	3.289	3.310	2.53×	21.12%	0.8053	0.6773	0.6235	0.8235
CUDA (Tesla T4)	GPU	0.392	—	—	—	0.8293	0.6832	0.6661	0.8301

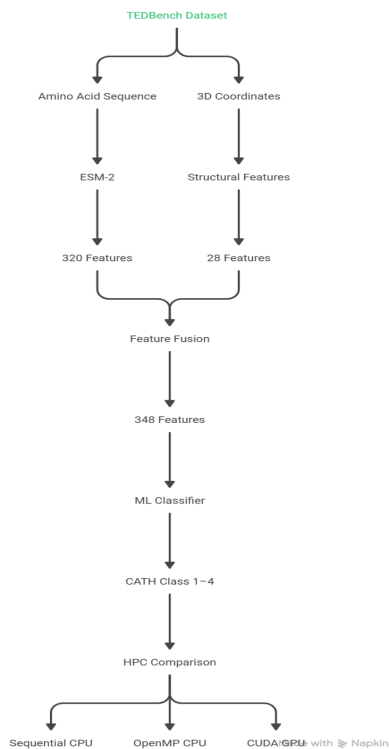
Feature Extraction Results

Platform	ESM Extraction (s)	Structural Extraction (s)	Total Pipeline (s)
CPU	1094.67	280.82	—
CUDA	78.766	439.152	518.31

ESM extraction speedup: $1094.67 / 78.766 \approx 13.9\times$

Project Structure

TEDBench Dataset Processing and HPC Comparison



Conclusion

The project combines protein sequence and 3D structural information for CATH Class classification.

ESM-2 generates 320 sequence features, while structural processing generates 28 structural features, giving 348 combined features for machine learning.

For the HPC component, OpenMP achieved a maximum $2.77\times$ speedup with 8 threads for the classification stage.

The CUDA experiment using a Tesla T4 achieved 82.93% test accuracy and reduced ESM feature extraction from 1094.67 s on CPU to 78.766 s on CUDA, approximately $13.9\times$ faster.

The current CUDA pipeline's largest time-consuming stage is structural feature extraction, which took 439.152 s.